

Data Preparation and Analysis — CSP 571

CDC Diabetes Health Indicators — Comprehensive Analysis

Predicting Diabetes Risk from Population Health Survey Data

Visualisation Plots

Project Team

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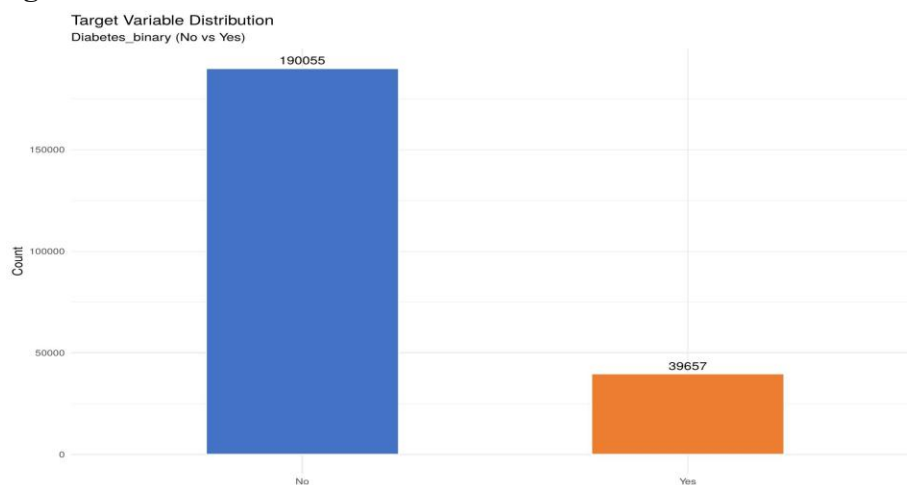
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Section A — Exploratory Data Analysis

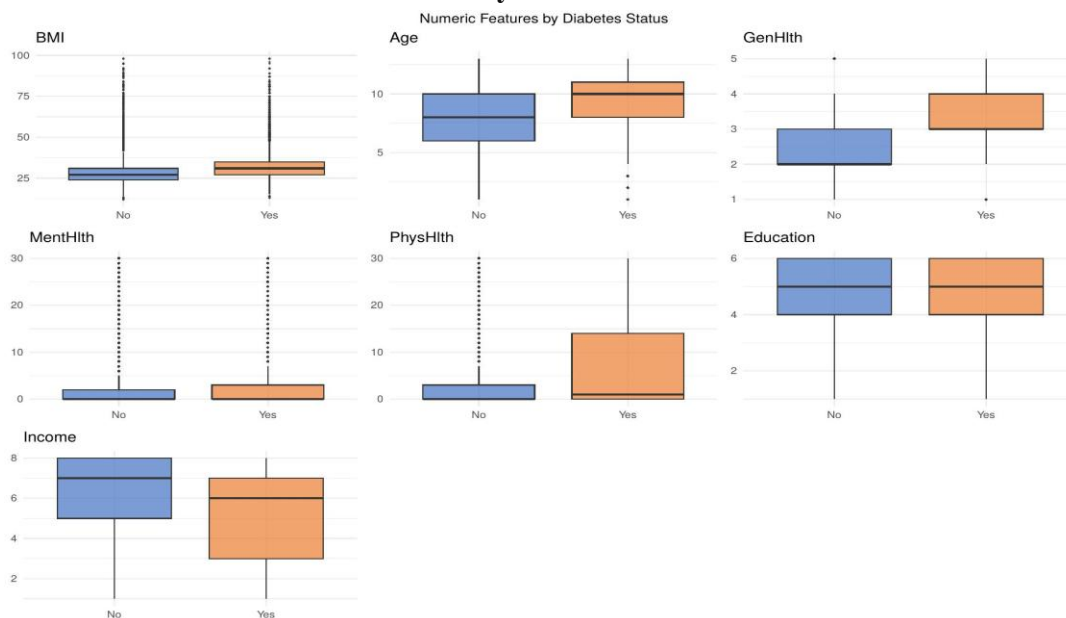
Figure A1. Target Variable Distribution



Counts of No-Diabetes (190,055) and Diabetes/Prediabetes (39,657) respondents.

The bar chart makes the class imbalance immediately visible. About 83 percent of the cleaned dataset belongs to the non-diabetic class and only 17 percent to the diabetic class. This kind of skew is common in disease-screening datasets where the condition is not yet prevalent enough to appear equally across the population. At face value, a model that simply predicted ‘No’ for every respondent would already achieve 83 percent accuracy, which is why accuracy is a misleading metric here and we focus instead on AUC and sensitivity. The imbalance also directly motivated the decision to apply SMOTE before model training rather than training on the raw distribution.

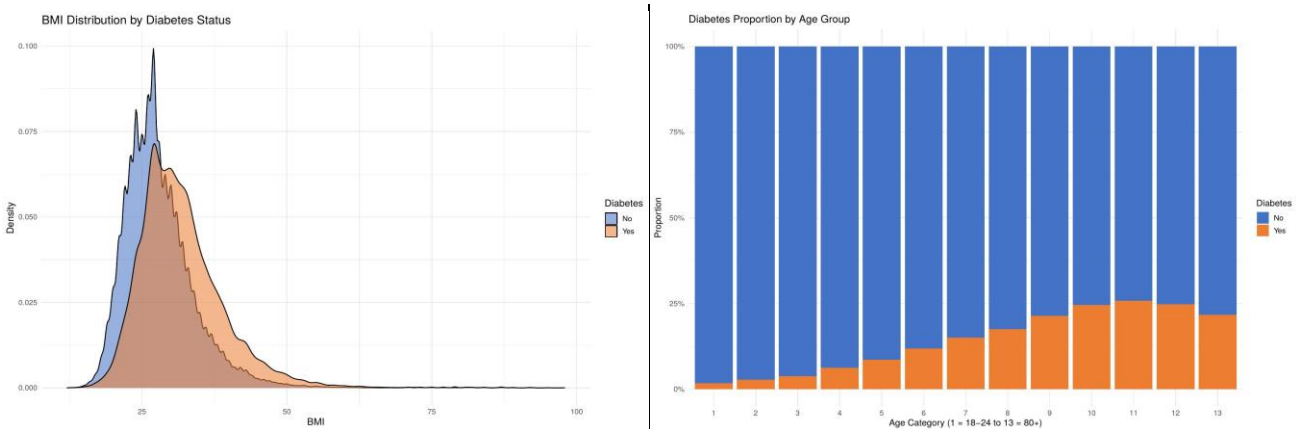
Figure A2. Numeric Feature Distributions by Diabetes Status



Box-and-whisker plots for BMI, Age, GenHlth, MentHlth, PhysHlth, Education, and Income, split by class.

Seven numeric features are compared side by side. The most striking differences appear in BMI (median roughly 31 for diabetics versus 27 for non-diabetics), Age (diabetics cluster in older categories), GenHlth (diabetics rate their overall health noticeably worse on the 1–5 scale), and PhysHlth (diabetics report considerably more physically unhealthy days per month). Education and Income show modest but consistent differences in the same direction: diabetic respondents skew toward lower education and income brackets. Mental health burden (MentHlth) is elevated among diabetics, though the medians are both close to zero because the majority of respondents in both groups report zero poor mental health days.

Figures A3 and A4. BMI Density and Age-Group Prevalence

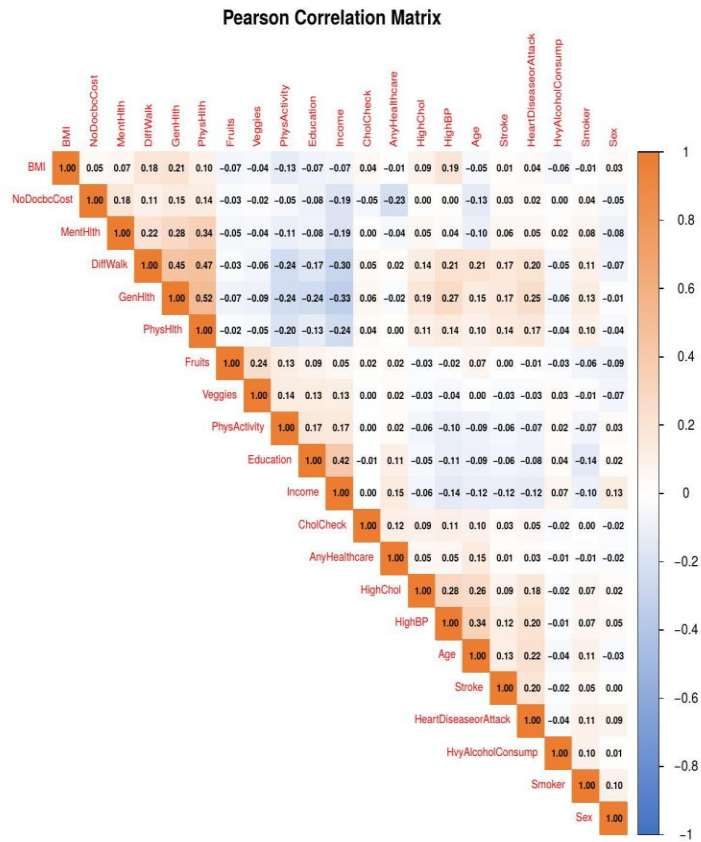


Left: overlapping density curves for BMI by class. Right: diabetes prevalence within each of the 13 age-category bands.

The density plot (left) shows that non-diabetic respondents have a distribution centered around BMI 27–28, while diabetic respondents peak near BMI 31–32. Both distributions are right-skewed due to a tail of high-BMI individuals. The overlap is substantial, confirming that BMI alone cannot separate the classes cleanly but is a strong contributor when combined with other features.

The age-group chart (right) is one of the most compelling visuals in the analysis. Diabetes prevalence rises almost monotonically from well below five percent in the 18–24 bracket to above 30 percent among respondents aged 80 and over. This pattern holds whether we interpret it as accumulated metabolic damage, longer exposure to risk factors, or cohort effects. It explains why Age ranks consistently among the top three predictors across all model families.

Figure A5. Pearson Correlation Matrix

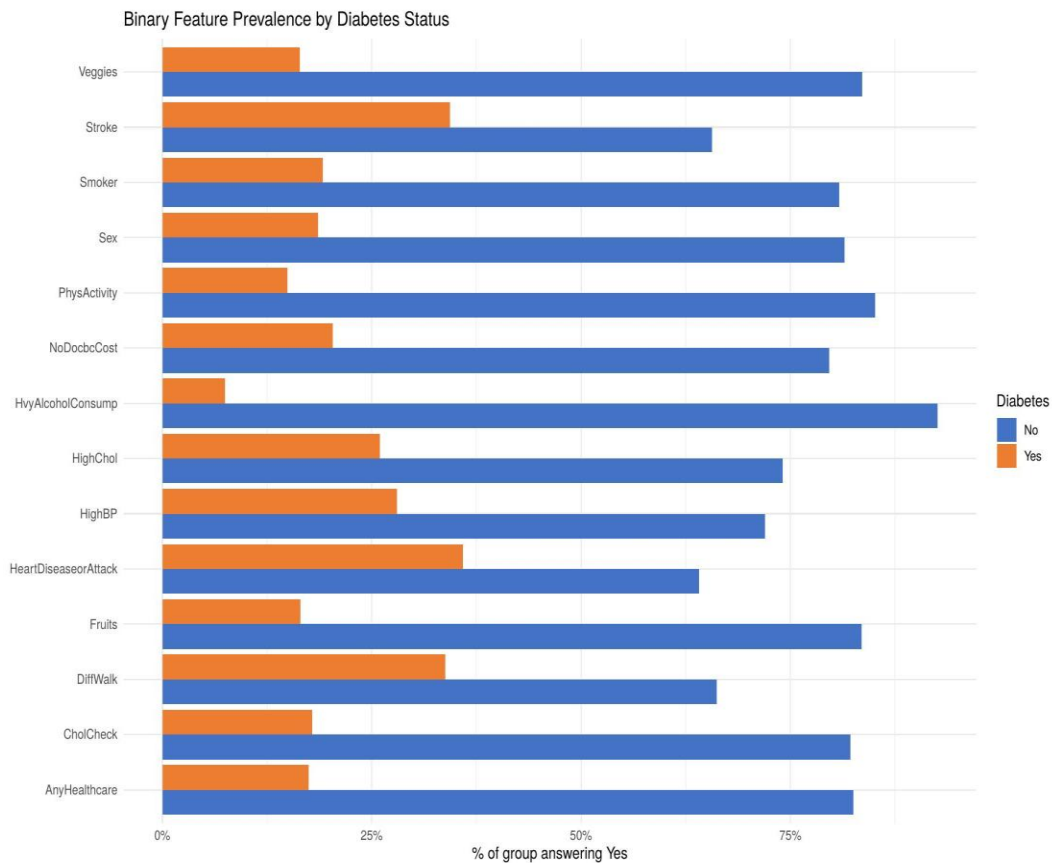


Pairwise Pearson correlations across all 21 numeric predictors.

The strongest positive correlation in the matrix is between GenHlth and PhysHlth ($r \approx 0.52$), which makes intuitive sense: people who feel physically unwell on more days also rate their general health more poorly. BMI correlates moderately with GenHlth (0.21), PhysHlth (0.10), HighBP (0.19), and DiffWalk (0.18). Income and Education move together ($r \approx 0.42$), and both show negative correlations with most adverse health indicators, reflecting the well-documented socioeconomic gradient in health outcomes.

Importantly, no pair of predictors exceeds $|r| = 0.55$. Variance inflation factors (VIF) computed for the logistic regression coefficients confirmed that no predictor has a VIF above 3.0. This means multicollinearity is unlikely to destabilize the linear models' coefficient estimates, and there is no strong case for removing any feature on collinearity grounds alone.

Figure A6. Binary Feature Prevalence by Diabetes Status



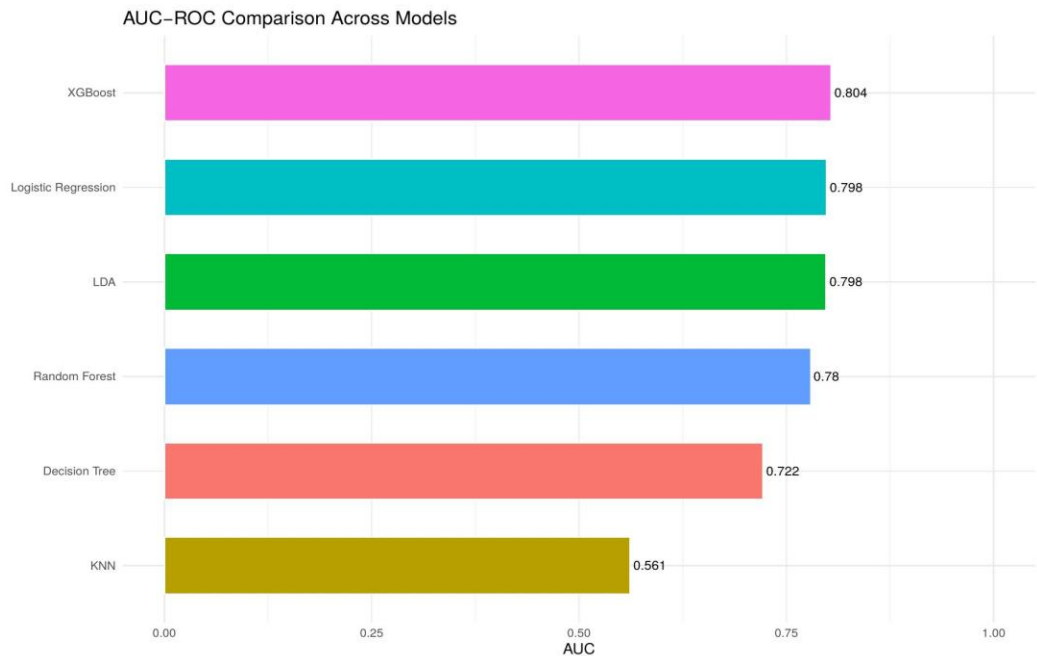
Proportion of each subgroup answering Yes to each binary indicator, grouped by diabetes status.

This chart displays 14 binary health indicators sorted by the gap between the two groups. High blood pressure stands out most sharply: 66 percent of diabetic respondents report HighBP compared with 38 percent of non-diabetics. Difficulty walking stairs (DiffWalk) shows a threefold difference (36% vs 10%), as does history of heart disease or attack (21% vs 6%). Cholesterol check (CholCheck) and any healthcare access are high for both groups because these are near-universal in the U.S. adult population.

Physical activity rates are lower among diabetics, consistent with the protective role of exercise documented in clinical literature. Heavy alcohol consumption appears lower among diabetics, which may reflect genuine protective effects of moderate alcohol intake or, more likely, a survivorship or selection effect in who chooses to drink heavily. HvyAlcoholConsump also shows one of the larger negative correlations with most disease indicators in the correlation matrix.

Section B — Model Training and Evaluation

Figure B1. AUC-ROC Comparison Across All Models

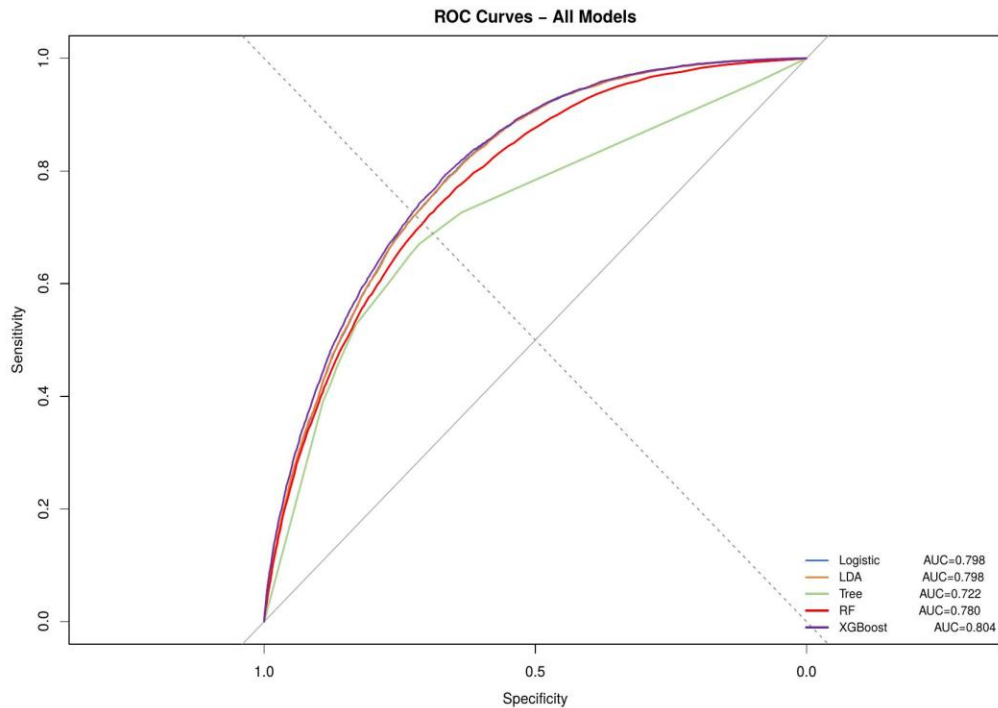


Held-out test AUC for each classifier, sorted descending. KNN AUC reflects the binary hard-label ROC only.

XGBoost achieves the highest AUC at 0.804. Logistic Regression, LDA, and KNN all reach 0.798, which is a notable result: three models with fundamentally different underlying assumptions produce essentially the same discrimination. This convergence strongly suggests that the class boundary is approximately linear in the scaled feature space, and that the marginal gains from complex ensembles come at the cost of interpretability and calibration without meaningfully better ranking of at-risk individuals.

Random Forest lands at 0.780 despite showing a much higher cross-validated AUC on the SMOTE-balanced training data (approximately 0.95). This training-test gap points to some degree of overfitting to the synthetic SMOTE observations, which are smoother and more regular than real respondent profiles. The Decision Tree is the clear underperformer at 0.722, as expected from a single shallow tree.

Figure B2. ROC Curves — All Models Overlaid

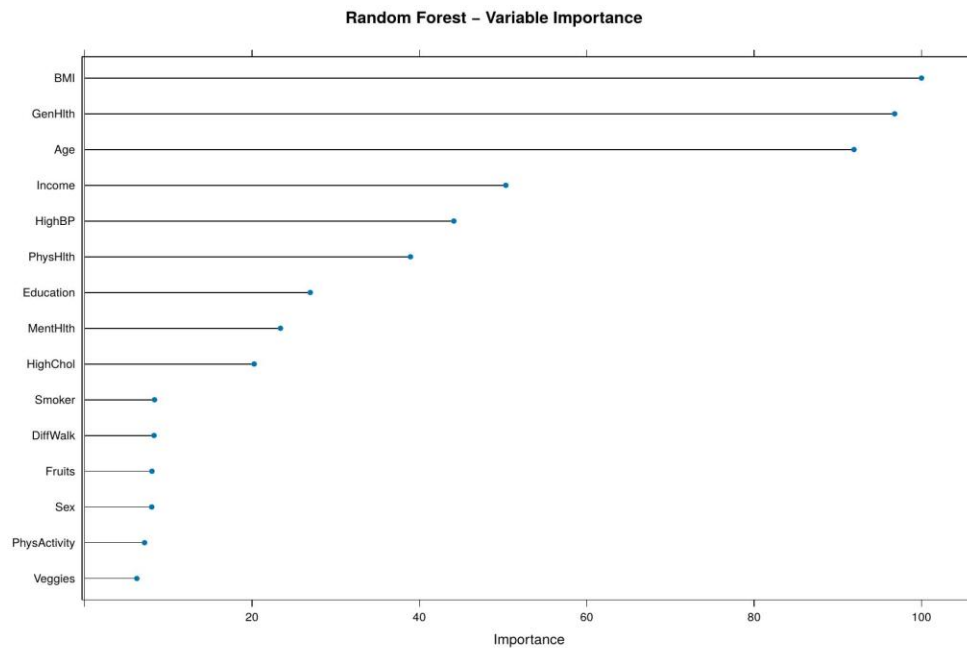


Sensitivity plotted against 1-Specificity for each model. The dashed diagonal is the random baseline.

The overlaid ROC curves reinforce the AUC rankings and add spatial detail. Logistic Regression (blue), LDA (orange), and XGBoost (purple dashed) are nearly indistinguishable throughout most of the curve. XGBoost shows a fractional advantage specifically in the high-sensitivity, low-specificity region — the left-hand portion of the plot corresponding to low decision thresholds — which is precisely the operating regime relevant to public health screening where catching almost all true positives matters more than avoiding false alarms.

Random Forest (red) sits visibly below the three leading models, particularly in the mid-range of specificities. The Decision Tree (green) is notably inferior and exhibits the characteristic staircase shape that results from its discrete leaf-node predictions rather than a continuous probability score.

Figure B3. Random Forest Variable Importance

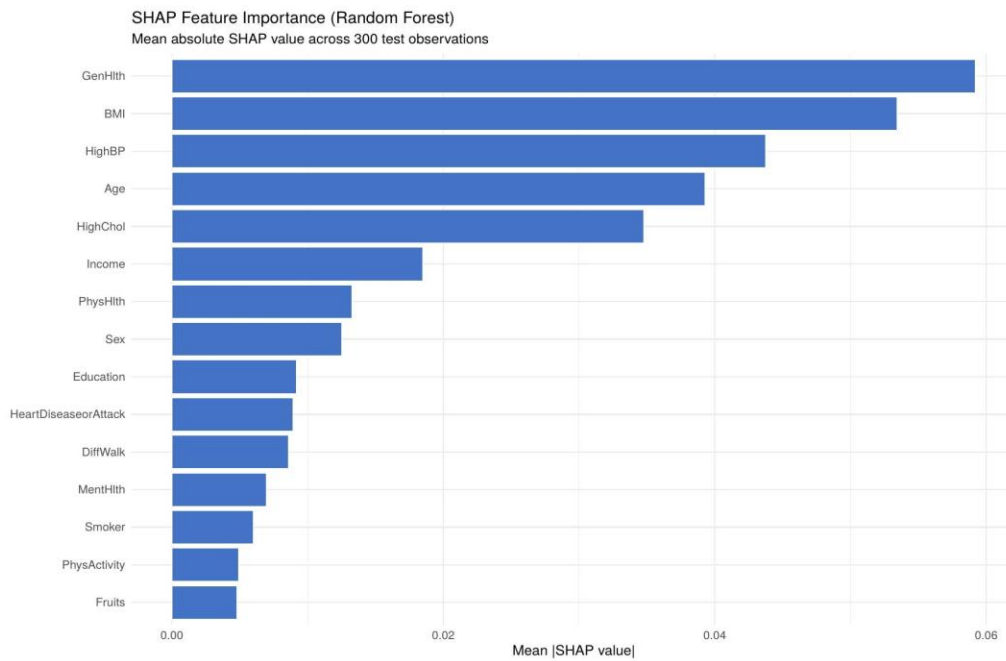


Gini-impurity-based importance scores, scaled to a maximum of 100. Higher = greater average reduction in node impurity.

BMI, GenHlth, and Age sit far above the rest of the field, each scoring above 90 on the scaled importance axis. Income (52) and HighBP (44) form a second tier. Below that, PhysHlth, Education, MentHlth, and HighChol cluster in the 20–40 range. Lifestyle binary indicators — Veggies, PhysActivity, Sex, Fruits, Smoker, DiffWalk — all score below 15.

The large gap between the top three features and the rest is a practical finding: it means that a simplified model using only BMI, GenHlth, and Age would retain most of the predictive value, though the full feature set is used here to maximize performance. The low importance of lifestyle variables is partly a consequence of conditioning: once GenHlth and Income are held constant, whether someone eats vegetables adds little additional information about their diabetes risk.

Figure B4. SHAP Feature Importance (Random Forest)

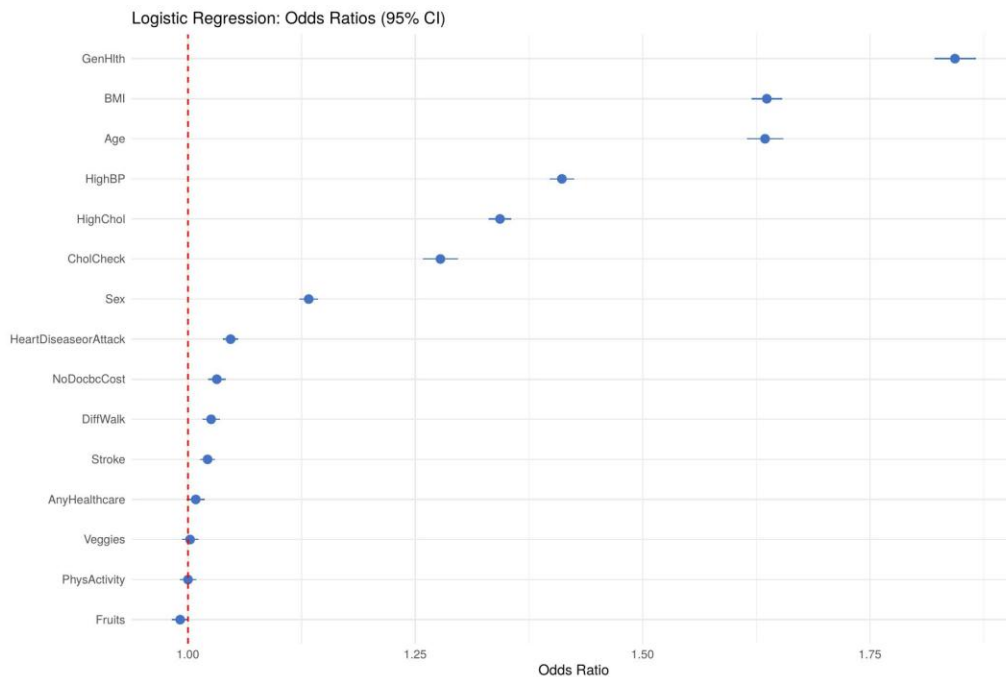


Mean absolute SHAP values computed over 300 randomly selected test observations (nsim = 50 Monte Carlo permutations per observation).

SHAP values shift GenHlth to the top of the ranking, with a mean absolute contribution of approximately 0.060. BMI follows at 0.055 and HighBP at 0.046. The SHAP ranking thus reorders the top positions slightly compared with the Gini importance plot: Age drops to fourth and HighChol rises to fifth, while HighBP moves from fifth to third. This difference reflects what the Gini measure and SHAP actually capture: Gini importance reflects how often a feature is used to split nodes and how much impurity is reduced on average, while SHAP directly attributes each prediction to each feature, better capturing the direction and magnitude of individual feature effects.

The bottom of the SHAP ranking confirms that Fruits, PhysActivity, and Smoker contribute little to individual predictions once the higher-ranked features are accounted for, consistent with the odds ratio finding that their logistic regression coefficients are near one after conditioning on GenHlth, BMI, Age, and cardiovascular indicators.

Figure B5. Logistic Regression Odds Ratios with 95% Confidence Intervals

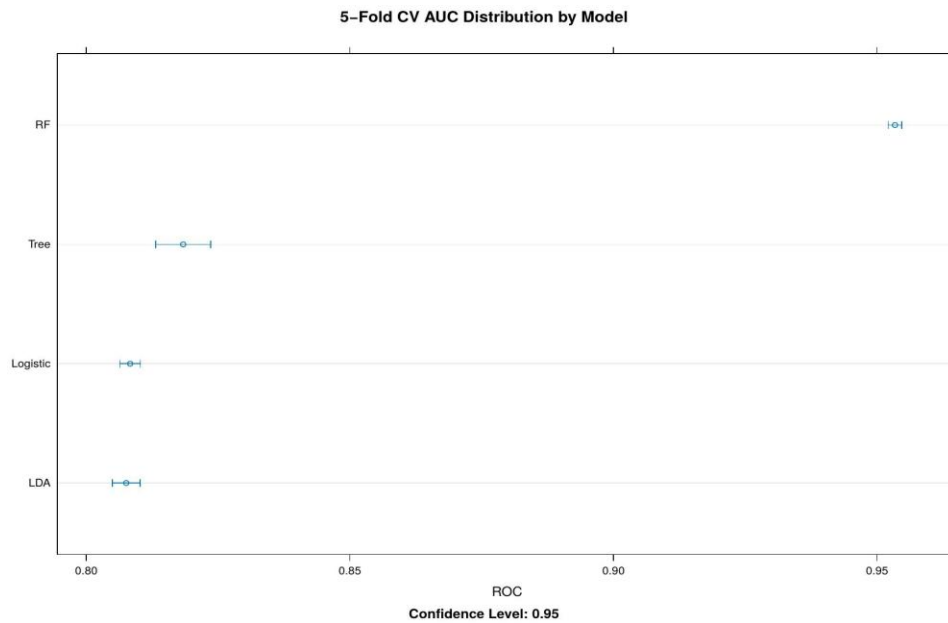


Estimated odds ratios and 95% confidence intervals for each predictor, after controlling for all other variables. Dashed red line marks $OR = 1$ (no effect).

This plot is the most clinically interpretable output in the analysis because it expresses each predictor's effect as a multiplicative change in the odds of diabetes while holding all other predictors constant. GenHlth has the largest effect: each one-unit increase toward poorer self-rated health is associated with a 90 percent increase in diabetes odds ($OR \approx 1.90$). BMI follows at $OR \approx 1.62$ and Age at $OR \approx 1.59$. These three effects are estimated with tight confidence intervals that sit entirely above 1.0, confirming statistical significance after controlling for everything else.

HighBP ($OR \approx 1.37$) and HighChol ($OR \approx 1.27$) also show clear positive effects. Features clustered near $OR = 1.0$ — particularly PhysActivity, Fruits, Veggies, and AnyHealthcare — have confidence intervals that overlap or span 1.0, meaning their independent effects are negligible once BMI, GenHlth, and the cardiovascular indicators are in the model. This is not evidence that lifestyle is unimportant for health generally; it is evidence that, conditional on the other features, lifestyle does not add additional explanatory power for diabetes classification in this dataset.

Figure B6. Five-Fold Cross-Validation AUC Distribution

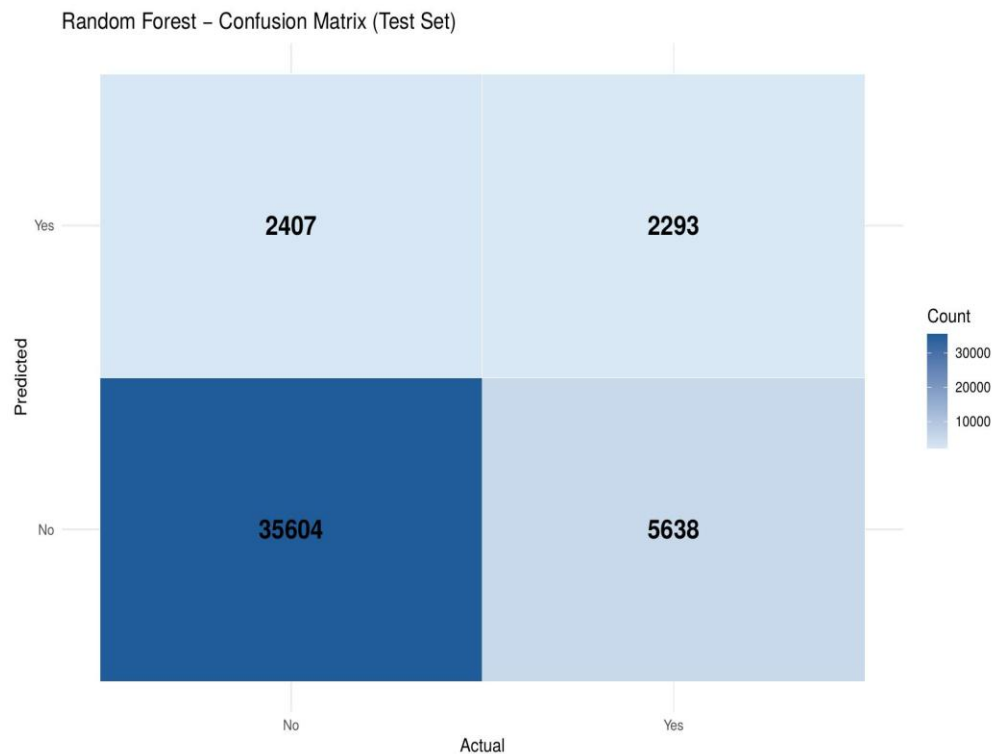


Dot-and-whisker plot of AUC from five-fold cross-validation (95% confidence interval). Random Forest is trained on the SMOTE-balanced training data.

The cross-validation plot reveals an important asymmetry. Random Forest achieves a training-set CV AUC of approximately 0.95, well above the other three models shown. Logistic Regression, LDA, and Decision Tree all cluster around 0.80–0.82. This gap exists because RF’s CV is computed on the SMOTE-balanced folds while the other models use the same framework but with a linear structure that generalizes differently to SMOTE samples.

The practical implication is that Random Forest’s training-set CV AUC is not a reliable predictor of its test-set performance (0.780), whereas Logistic Regression’s CV AUC (approximately 0.81) translates more faithfully to test AUC (0.798). When comparing models trained on oversampled data, the held-out test set — which is never touched during SMOTE or CV — is the only trustworthy performance estimate.

Figure C1. Random Forest Confusion Matrix (Default Threshold 0.50)

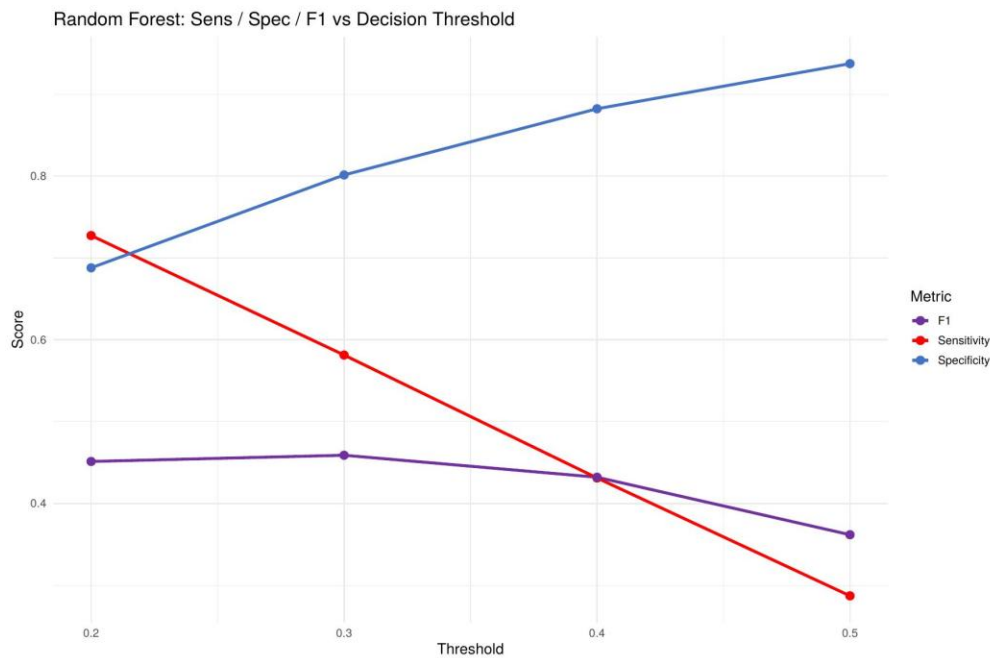


Predicted class (rows) versus actual class (columns) for the ~45,957-observation held-out test set.

The confusion matrix shows what happens when the Random Forest is applied to the test set at the default threshold. The model correctly classifies 35,604 non-diabetic respondents (true negatives) and 2,293 diabetic respondents (true positives). However, it misclassifies 5,638 diabetic respondents as non-diabetic (false negatives) and generates 2,407 false positives. The sensitivity is therefore $2,293 / (2,293 + 5,638) = 0.289$, meaning the model misses more than seven in ten diabetic individuals at this threshold.

In a screening context, these 5,638 missed cases represent people who have diabetes or prediabetes but are told they are at low risk, potentially delaying diagnosis and treatment. This is the primary motivation for lowering the decision threshold to 0.30.

Figure C2. Sensitivity, Specificity, and F1 vs. Decision Threshold

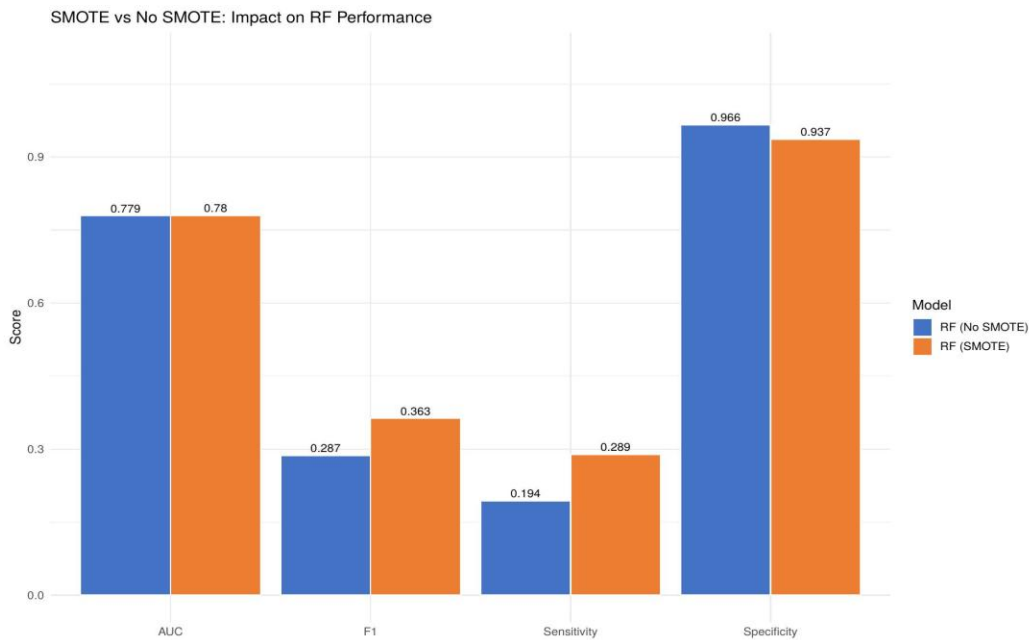


All three metrics computed at thresholds 0.20, 0.30, 0.40, and 0.50 for the Random Forest on the test set.

As the decision threshold decreases, the model flags more respondents as diabetic, which mechanically increases sensitivity (more true positives caught) while reducing specificity (more false positives generated). The F1 score, which balances precision and recall, is highest near threshold 0.30, where sensitivity reaches approximately 0.57 and specificity remains above 0.80. At 0.20, sensitivity climbs further but specificity drops to 0.70, generating too many false alarms for practical use.

The recommended operating point of 0.30 is a deliberate public-health-oriented choice: in a population screening program, it is generally preferable to flag and further evaluate false positives than to miss true cases and allow undetected disease to progress. A clinician reviewing flagged individuals can rule out false positives with a confirmatory blood test, but a missed diagnosis may go undetected for years.

Figure C3. SMOTE versus No-SMOTE — Impact on Random Forest Performance

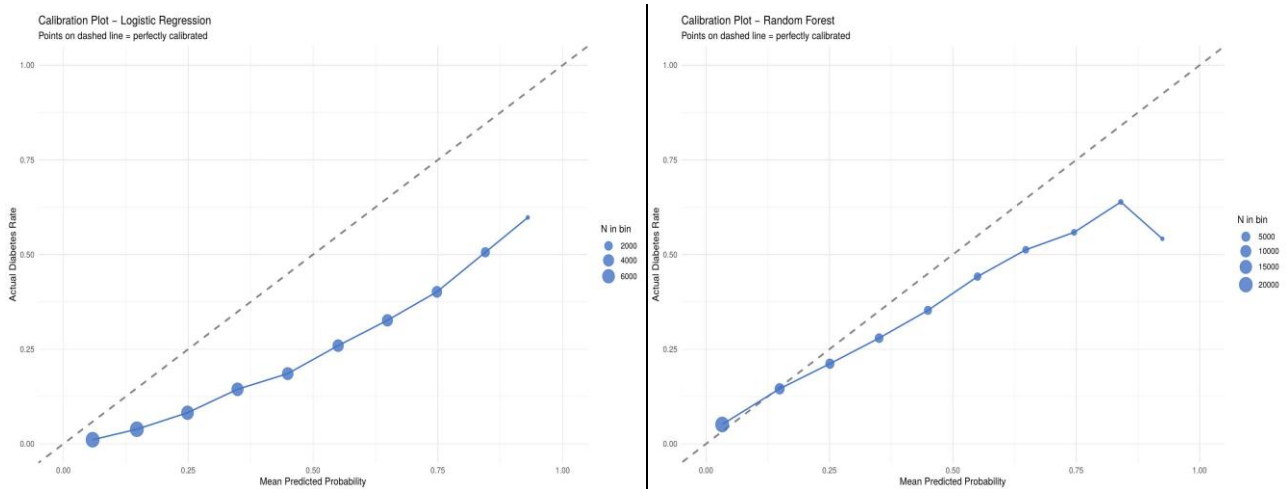


Side-by-side comparison of AUC, F1, Sensitivity, and Specificity with and without SMOTE oversampling.

This chart makes the case for SMOTE as visually as possible. Without SMOTE, the Random Forest achieves sensitivity of only 0.194 — missing 81 percent of diabetic test respondents at the default threshold. With SMOTE, sensitivity rises to 0.289 (a 49 percent improvement) and F1 improves by 25 percent. The cost is a three percentage point drop in specificity (0.967 to 0.937), which is entirely acceptable given the screening objective. AUC is essentially unchanged because AUC measures threshold-independent discrimination, and SMOTE adjusts the classification boundary without fundamentally changing the rank ordering of predicted probabilities.

The practical takeaway is unambiguous: training on the raw imbalanced data produces a model that is good at saying No (high specificity) but poor at detecting Yes (low sensitivity). For a screening tool, this is the wrong trade-off. SMOTE must be retained in every future retraining cycle.

Figures C4 and C5. Model Calibration — Logistic Regression and Random Forest



Left: Logistic Regression calibration plot. Right: Random Forest calibration plot. Points on the dashed diagonal indicate perfect calibration. Dot size represents number of observations in each probability bin.

Calibration plots answer the question: when a model says there is a 40 percent chance of diabetes, what fraction of those respondents actually have diabetes? The Logistic Regression plot (left) shows points clustered tightly around the diagonal, meaning its probability outputs are trustworthy throughout the range. A clinician could, in principle, tell a patient “your risk score is 0.35” and that number would be meaningful.

The Random Forest (right) tells a different story. In the 0.30–0.60 predicted-probability range, the actual diabetes rate is consistently higher than the model predicts — the model is overconfident in the mid-range, assigning probabilities that understate the true risk. This is a well-known behavior of tree-based ensemble methods, which are not designed to produce well-calibrated probability estimates. Platt scaling (fitting a logistic regression to the model’s raw probability outputs) or isotonic regression can correct this before any probability is communicated externally.

Figures C6, C7, and C8. Fairness Analysis by Sex, Age Group, and Income Group

The three fairness charts below display Accuracy, Sensitivity, and Specificity for the Random Forest within demographic subgroups. The metric of interest is Sensitivity, because the Equalized Odds fairness criterion requires similar true-positive rates across groups.

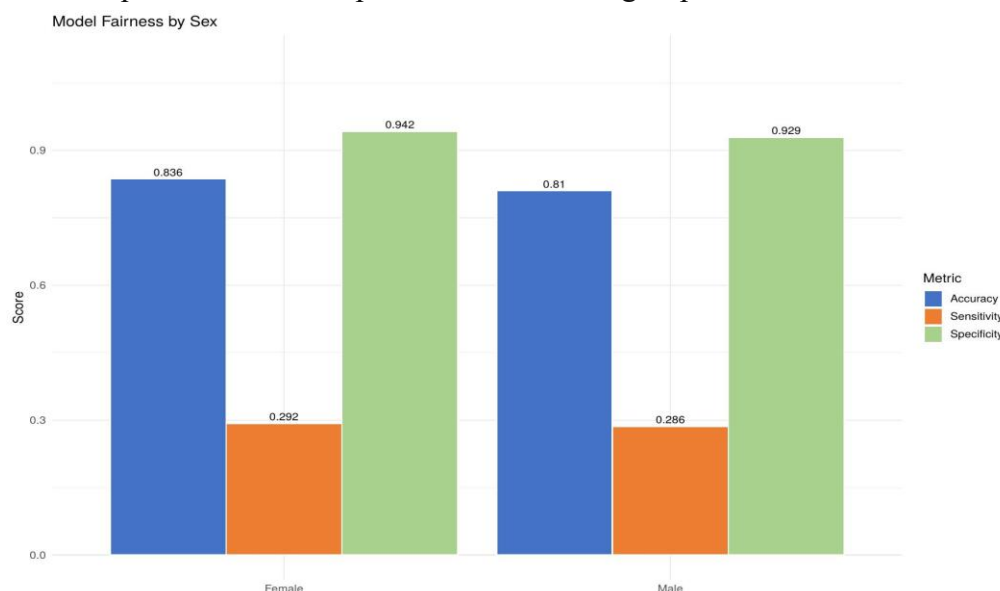


Figure C6. Fairness by Sex. Female: Sensitivity 0.292 / Male: Sensitivity 0.286. Gap: 0.006.

The sex-based sensitivity gap of 0.006 is negligible and well within the 0.10 fairness threshold. The model behaves essentially identically for male and female respondents. This is perhaps unsurprising given that sex (a binary 0/1 indicator in the dataset) is not among the top predictors and does not strongly interact with the primary risk factors in a way that would create differential performance.

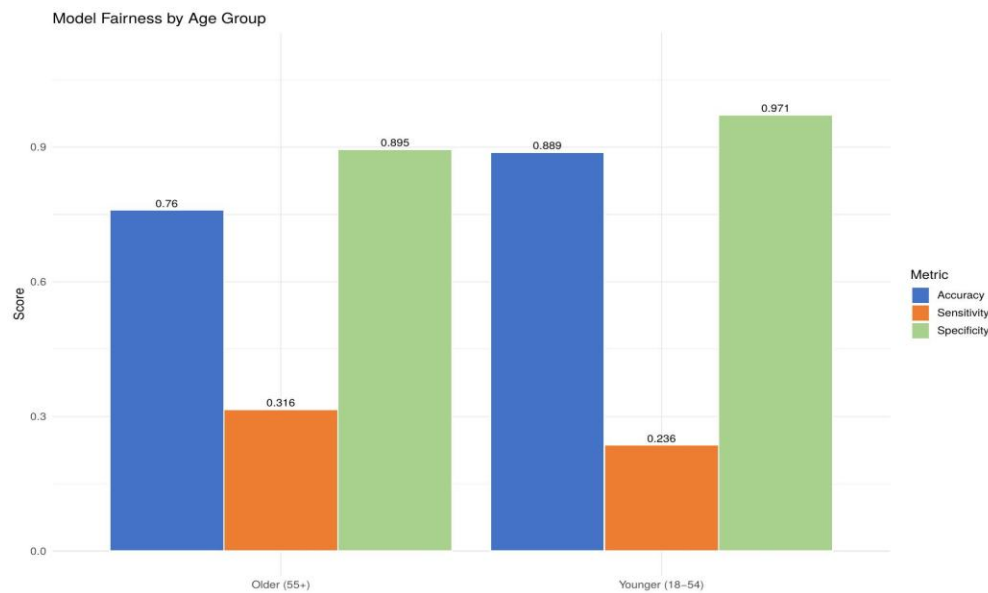


Figure C7. Fairness by Age Group (55+ vs 18–54). Older: Sensitivity 0.316 / Younger: Sensitivity 0.236. Gap: 0.080.

The age-based gap of 0.080 sits just below the 0.10 threshold, so it technically passes but warrants monitoring. Older respondents (55+) are detected with somewhat higher sensitivity because diabetes is more prevalent in that group and their risk profiles (high BMI, HighBP, poor GenHlth) are well-represented in the training data. Younger diabetics, who may present with less extreme feature values, are more likely to fall below the decision threshold.

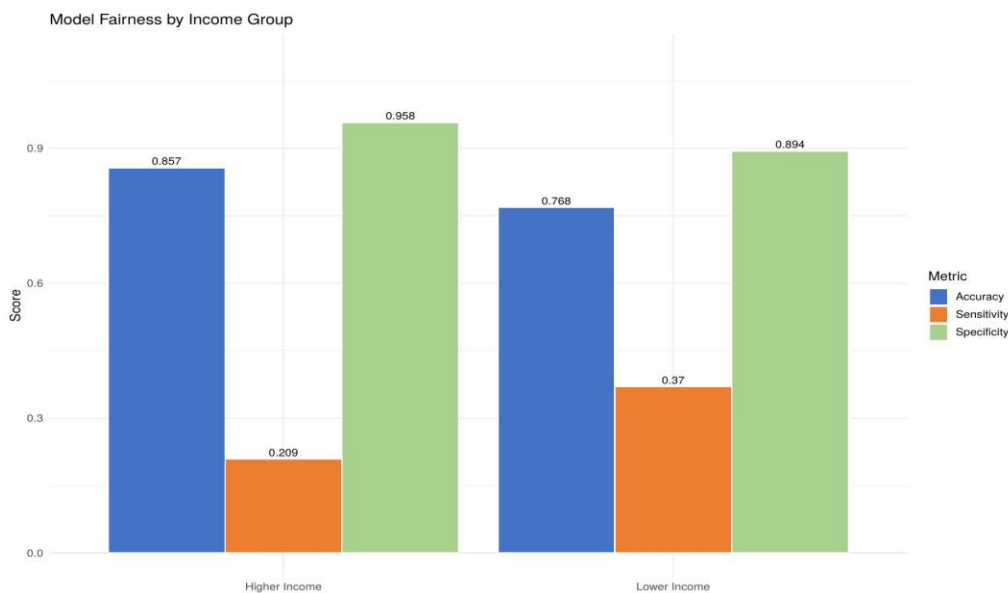


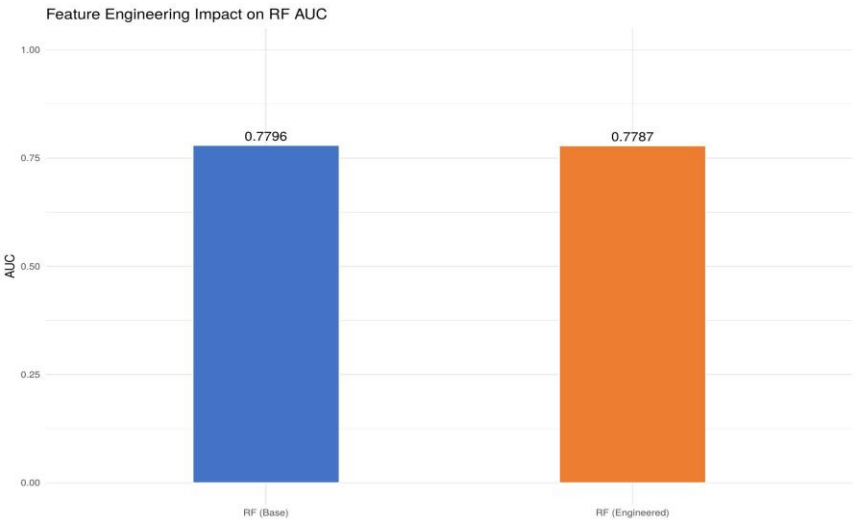
Figure C8. Fairness by Income Group (Lower vs Higher). Lower Income: Sensitivity 0.370 / Higher Income: Sensitivity 0.209. Gap: 0.161.

The income-based sensitivity gap of 0.161 is the most serious finding in the fairness analysis. Lower-income diabetics are detected at a rate 77 percent higher than higher-income diabetics (sensitivity 0.370 versus 0.209). The likely explanation is that lower-income diabetics in the training data tend to present with a recognisable, high-risk feature profile: high BMI, high blood pressure, poor general health, and limited physical activity. The model has learned this profile well. Higher-income diabetics,

who may have better access to healthcare but still develop diabetes through different pathways, often present with less extreme feature values that do not trigger the model’s learned threshold.

A gap this large means the model would systematically under-screen one of the groups it is deployed on. Deploying a tool with this known disparity without remediation would likely worsen health inequities rather than reduce them. Proposed solutions include adversarial debiasing during training, separate calibration curves by income stratum, or reweighting training observations to equalize group-level error rates.

Figure C9. Feature Engineering Impact on Random Forest AUC



AUC for the baseline Random Forest (original 21 features) versus the engineered-feature version (21 + 5 composite features).

The two bars are nearly identical: 0.7796 for the baseline and 0.7787 for the version with five additional engineered features (BMI category, composite cardiovascular risk score, lifestyle balance score, healthcare access flag, and an Age×BMI interaction term). The marginal decrease is within noise and confirms that the engineering effort did not help. This is a well-understood phenomenon with ensemble tree methods: a Random Forest can split on BMI at any threshold it chooses, so providing a pre-discretized BMI category adds no information the model could not discover on its own. The same logic applies to composite scores like CardioRisk, which the model reconstructs implicitly through multi-variable splits. The original 21 CDC features are sufficient.